Morphophonology

Class 2: Boundary effects without boundaries

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Summary of last time

- Observation: phonology may apply unexpectedly in morphologically complex words
- Hypothesis: morphological boundaries can be mirrored by prosodic boundaries
- ALIGN constraints favor prosodic boundaries at morpheme boundaries
- May cause phonology to apply differently

Morpheme boundaries as prosodic boundaries

- Prosodic approach directly encodes the effect of morpheme boundaries by making phonology sensitive to boundaries
- ¿But why should prosody care so much about morpheme boundaries?
- ¿And why should phonological processes care so much about prosodic boundaries?
- ¿Are prosodic boundaries a *cause* or a *consequence* of how phonology applies to morphologically complex forms?

Prosodic boundaries: a "cheap fix"?

- German [aʊfʔɛsən]
  - [aʊf.ʔɛ.sən]: morpheme and syllable boundaries do not actually(?) align
- Granada Spanish [dehetʃo] *deshecho* 'undone'
  - [de.he.tʃo]: morpheme and syllable boundaries do not actually align

An alternative

- Morphological structure influences phonology indirectly
- Various possible mechanisms
  - Morphology restricts the domains that phonology evaluates
  - Morphology establishes correspondence relations between expressions that share morphological structure

Three approaches to compare

- Aligned prosodic and morpheme boundaries
    - Prosodic domains (last time)
- Cyclic evaluation (with input-output faithfulness)
- Transderivational correspondence (with output-output faithfulness)

Cyclic Evaluation

- Interleaving of morphology and phonology
- Expressions are evaluated when additional morphology is added
  - Numerous possible versions have been explored
- Important consequence: some phonological evaluations take place before the entire expression has been composed

Korean cluster simplification

- Korean phonotactics ban final CC#
  - [tʃip] 'house', [sok] 'inside', [nun] 'eye', [mal] 'speech'
  - *[palk], *[naps], *[talm], etc.
- Cluster simplification

/kaps/	kap	'price'
/hɨlk/	hɨk	'soil'
/talk/	tak	'chicken'
/salm/	sam	'life'

- Medial clusters are allowed
  - [əlgul] 'face'

Cluster simplification in OT

- "Normal application": simplify in final position (only)

/talk/	*CC]	Max(C)	Dep(C)
a. ☞ tak		*	
b. talk	*!		

/əlkul/	*CC]	Max(C)	Dep(C)
a. ☞ əlgul			
b. əgul		*	

Overapplication of cluster simplification

Contemporary Korean (Ko 2006, and many others)

/kaps-i/	kabi	*kapsi
/hɨlk-i/	hɨgi	*hɨlgi
/talk-i/	tagi	*talgi
/salm-i/	sami	*salmi

- The problem: why don't the medial clusters surface?
  - [talgi] would be perfectly legal (cf. [əlgul] 'face')

Cyclic evaluation of complex words

- Interleaving of morphology and phonology

Cycle 1	/talk/
Cluster simplification	tak
Cycle 2	/tak-i/
Cluster simplification	—
Output	tagi

- Cluster is simplified before context that could preserve it is added (counterbleeding of phonology by morphology)

Cyclic evaluation: in OT

- First cycle

/talk/	*CC]	Max(C)	Dep(C)
a. ☞ tak		*	
b. talk	*!		

- Second cycle

/tak-i/	*CC]	Max(C)	Dep(C)
a. ☞ taki			
b. talki			*!

Laryngeal neutralization

- Korean has a three-way laryngeal contrast

	Lax	Aspirated	Tense
p	발 pal 'foot'	팔 p^{h}al 'arm'	빨래 p*alːɛ 'laundry'
t	대 tɛ 'stalk'	테 t^{h}e 'frame'	때 t*ɛ 'time'
k	개 kɛ 'dog'	캐 k^{h}ɛ 'dig!'	깨 k*ɛ 'perilla'

- Only lax stops in final position

Lax	Aspirated	Tense
/sok/ sok̚ 'inside'	/puək^h/ puək̚ 'kitchen'	/pak*/ bak̚ 'outside'
/tʃip/ tʃip̚ 'house'	/sot^h/ sot̚ 'pot'	

Laryngeal neutralization: in OT

- Contrast in initial position

/te/	*Cʰ], C*]	Ident(c.g.)	Ident(s.g.)
a. ☞ te			
b. t^he			*

/t^he/	*Cʰ], C*]	Ident(c.g.)	Ident(s.g.)
a. te			*!
b. ☞ t^he			

/t*e/	*Cʰ], C*]	Ident(c.g.)	Ident(s.g.)
a. te		*!	
b. ☞ t*e			

Laryngeal neutralization: in OT

- Neutralized in final position (hypothetical morphemes)

/tak/	*Cʰ], C*]	Ident(c.g.)	Ident(s.g.)
a. ☞ tak			
b. takʰ	*!		*

/takʰ/	*Cʰ], C*]	Ident(c.g.)	Ident(s.g.)
a. ☞ tak			*
b. takʰ	*!		

/tak*/	*Cʰ], C*]	Ident(c.g.)	Ident(s.g.)
a. ☞ tak		*	
b. tak*	*!		

Laryngeal neutralization: in OT

- Intervocalic position: contrast, but with voicing

/taka/	$*D^h$, *D*	$*C^h$], C*]	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a. taka						*!	
b. tak^ha				*!	*	*	
c. tak*a			*!			*	
d. ☞ taga							*

No voicing of aspirated, tense stops

/tak^{h}a/	*D^h, *D*	*C^h], C*]	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a. taka				*!		*	
b. ☞ tak^{h}a					*	*	
c. tag^{h}a	*!				*		*
d. taga				*!			*

/tak*a/	*D^h, *D*	*C^h], C*]	Ident(c.g.)	Ident(s.g.)	*c.g.	*V[-voi]V	Ident(voi)
a. taka			*!			*	
b. ☞ tak*a					*	*	
c. tag*a	*!						*
d. taga			*!				*

Overapplication of laryngeal neutralization

Contemporary Korean (Ko 2006, and many others)

/sup^h-i/	subi	'forest-NOM'
/cip^h-ɨl/	cibɨl	'straw-ACC'
/mulip^h-e/	muribe	'knee-LOC'
/puək^h-ɨn/	puəgɨn	kitchen-TOP'

- The problem: why don't underlyingly aspirated C^h surface faithfully before suffix?
- [sup^{h}i] would be phonotactically fine

Overapplication of laryngeal neutralization

- Cycle 1

/sup^h/	*D^h, *D*	*C^h], C*]	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a. ☞ sup				*			
b. sup^h		*!			*		

- Cycle 2

/sup-i/	*D^h, *D*	*C^h], C*]	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a. ☞ subi							*
b. supi						*!	
c. sup^{h}i				*!	*	*	
d. sub^{h}i	*!			*	*		*

Summary of Korean overapplication

- Neutralization process applies in first cycle
- Neutralized output is carried forward into cycle 2
- Result: overapplication of neutralization in suffixed form

English aspiration

- Recall misapplication of aspiration after prefixes

dis[p]lay	dis[pʰ]lease
des[p]otism	dis[pʰ]ossess
whis[p]ering	mis[pʰ]erceive
pis[t]achio	mis[tʰ]abulate
Pis[k]ataway	mis[kʰ]ategorize

- Cyclic approach: stage of evaluation in which stem-initial stops are word-initial

Cyclic misapplication of aspiration?

- Cycle 1

/pliːz/	*sT^h	T^h/#_V	*C^h	IO-Ident([±asp])
a. ☞ p^{h}liz			*	*
b. pliːz		*!		

- Cycle 2: a problem!

/dɪs-p^{h}liːz/	*sT^h	T^h/#_V	*C^h	IO-Ident([±asp])
a. 😿 dɪsp^{h}liz	*!		*	
b. ☞ dɪspliːz				*

The problem

- Cycle 1: predictable initial aspiration
  - /pliːz/ → [pʰliːz]
- Cycle 2: prefix changes the context, we predict de-aspiration
  - /dɪs-pʰliːz/ → *[dɪspliːz]
- Diagnosis: because aspiration is allophonic in English, grammar both aspirates and deaspirates as needed
- How do we fix this?

The solution: re-ranking across cycles

- Cycle 1

/pliːz/	*sTʰ	Tʰ/#_V	*Cʰ	IO-Ident([±asp])
a. ☞ pʰliz			*	*
b. pliːz		*!		

- Cycle 2: preserve aspiration with higher Ident

/dɪs-pʰliːz/	IO-Ident([±asp])	*sTʰ	Tʰ/#_V	*Cʰ
a. ☞ dɪspʰliz		*		*
b. dɪspliːz	*!			

The logic of the approach

- Phonology assigns certain properties (feature values) at an earlier stage of the derivation, and those properties are then retained through later stages
- Assigning properties, in OT:
  - Markedness >> Faithfulness *for the relevant property*
- Retaining properties:
  - Faithfulness >> Markedness

Assigning properties in OT: $\mathcal{M} \gg \mathcal{F}$

Markedness constraints demand predictable changes to feature values, as required by context

/sa/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a. ☞ sa			
b. za		*!	*

/asa/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a. asa	*!		
b. ☞ aza		*	*

Assigning properties in OT: $\mathcal{M} \gg \mathcal{F}$

Predictably: even if input is specified wrongly for this feature, markedness demands the right output

/asa/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a. asa	*!		
b. ☞ aza		*	*

/aza/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a. asa	*!		*
b. ☞ aza		*	

When values are predictable…

/asa/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a. asa	*!		
b. ☞ aza		*	*

/aza/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a. asa	*!		*
b. ☞ aza		*	

Richness of the base (ROTB): analysis doesn't impose any requirements on UR's for this particular property, in this particular language

Retaining properties in OT: $\mathcal{F} \gg \mathcal{M}$

Faithfulness constraints demand that underlying feature values are preserved on the surface

/sa/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,-voi]
a. ☞ sa			
b. za	*!		*

/asa/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,-voi]
a. ☞ asa		*	
b. aza	*!		*

Retaining properties in OT: $\mathcal{F} \gg \mathcal{M}$

Faithfulness constraints demand that underlying feature values are preserved on the surface

/za/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,-voi]
a. sa	*!		
b. ☞ za			*

/aza/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,-voi]
a. asa	*!	*	
b. ☞ aza			*

Restating the logic of the approach, in OT

- Properties are assigned at earlier stages of derivation
  - Markedness ≫ Faithfulness
- Those properties are retained at later stages
  - Faithfulness ≫ Markedness (*reranking*)
- "Stratal" OT (Kiparsky, Bermúdez-Otero, and others)

/pliːz/	*sTʰ	Tʰ/#_V	*Cʰ	IO-Ident([±asp])
a. ☞ pʰliz			*	*
b. pliːz		*!		

/dɪs-pʰliːz/	IO-Ident([±asp])	*sTʰ	Tʰ/#_V	*Cʰ
a. ☞ dɪspʰliz		*		*
b. dɪspliːz	*!			

A cloud on the horizon: ROTB

- Re-ranking allows for contrast at later cycles
- Affixes introduced at later cycles are not subject to predictable allophonic repairs (Benua 1996)

/pliːz/	*sTʰ	Tʰ/#_V	*Cʰ	IO-Ident([±asp])
a. ☞ pʰliz			*	*
b. pliːz		*!		

/pɹiː-pʰliːz/	IO-Ident([±asp])	*sTʰ	Tʰ/#_V	*Cʰ
a. ☞ pɹiːpʰliz		*		*
b. 😿 pʰɹiːpliːz	*!			

(We'll consider some possible patches later)

Summary of cyclic derivation

- Phonological evaluations apply throughout the morphological derivation of words
    - "Misapplication" in complex words when there is an inner cycle
- Features assigned during earlier cycles may be preserved in subsequent cycles
    - Or, affect how other processes apply (opacity)
- A tension: preserving properties of stems, while allowing phonological processes to continue to apply to affixes

Transderivational correspondence

- A shared insight: cluster simplification in Korean /talk-i/ → [tagi] is due to the fact that it contains a subpart [tak]
- Difference: how that phonological evaluation references that subpart
- Cyclic faithfulness: output of earlier stage is input to the next stage
  - IO-Faithfulness to earlier stages of derivation
- Transderivational faithfulness: base is a separate word
  - OO-Faithfulness to related derivations

Cluster simplification in *tagi*

<u>Cyclic approach</u>

↓ *Morphology*

/talk/

↓ *Phonology*

[tak]

↓ *Morphology*

/tak-i/

↓ *Phonology*

[tagi]

<u>Transderivational approach</u>

↓	*Morphology*	↓
/talk/		/talk-i/
↓	*Phonology*	↓
[tak]	→	[tagi]

IO vs. OO Faithfulness

- Simple and complex expressions are each evaluated separately by the phonology
- Surface form of complex expressions constrained to resemble…
  - Their UR
  - Surface forms of related expressions (usually: base of affixation: Kenstowicz 1996, Benua 1996, Burzio 1996)
- Similarity is enforced by correspondence constraints (faithfulness; McCarthy and Prince 1995)

Overapplication of CC simplification with BD-Ident

- Ko (2006)

/talk/	BD-Dep(C)	*CC]	IO-Max(C)	IO-Dep(C)
a. ☞ tak			*	
b. talk		*!		

/talk-i/ Base: [tak]	BD-Dep(C)	*CC]	IO-Max(C)	IO-Dep(C)
a. ☞ tagi			*	
b. talgi	*!			

English: *dis*[p]*lay* vs. *dis*[pʰ]*lease*

/dɪspleɪ/	BD-Ident ([±asp])	*sCʰ	TʰV́	IO-Ident ([±asp])
a. ☞ dɪˈspleɪ			*	
b. dɪˈspʰleɪ		*!		*

/pliz/	BD-Ident ([±asp])	*sCʰ	TʰV́	IO-Ident ([±asp])
a. pliz			*!	
b. ☞ pʰliz				*

/dɪs-pliz/	BD-Ident ([±asp])	*sCʰ	TʰV́	IO-Ident ([±asp])
a. dɪspliz	*!		*	
b. ☞ dɪspʰliz		*		*

Aspiration in prefixes

/pliz/	BD-Ident([±asp])	*sC^h	$T^h\acute{V}$, #T^hV	IO-Ident([±asp])
a. pliz			*!	
b. ☞ p^hliz				*

/pɹi-pliz/	BD-Ident([±asp])	*sC^h	$T^h\acute{V}$, #T^hV	IO-Ident([±asp])
a. pɹipliz	*!		**	
b. pɹip^hliz			*!	*
c. ☞ p^hɹip^hliz				**

/p^hɹi-pliz/	BD-Ident([±asp])	*sC^h	$T^h\acute{V}$, #T^hV	IO-Ident([±asp])
a. pɹipliz	*!		**	*
b. pɹip^hliz			*!	**
c. ☞ p^hɹip^hliz				*

A cloud on the horizon

- Misapplication in [dɪsp^{h}liːz] relies on the existence of an independent word form [p^{h}liːz]
- Prediction: normal application with bound stems
- Potential problems…

expression	ɛkspɹɛʃn̩	*expropriate*	ɛksp^{h}ɹoʊpɹieɪt
expunge	ɛkspʌndʒ	*discordant*	dɪsk^{h}ɔɹdn̩t (?)
extenuating	ɛkstɛnjueɪɾɪŋ		
dispensary	dɪspɛnsəɹi		

Summary of transderivational faithfulness

- Regular application: ranking of Markedness and IO-Faithfulness
- Misapplication in complex forms may occur when there is a free-standing base form
  - Higher-ranked BD-Faithfulness
- Regular phonology still applies when it does not violate BD-Ident
  - e.g., in affixes

References

Benua, Laura. 1997. Transderivational Identity: Phonological Relations Between Words. UMass Amherst PhD.

Berry, Lynn. 1998. Alignment and Adjacency in Optimality Theory: evidence from Warlpiri and Arrernte. University of Sydney PhD.

Burzio, Luigi. 1996. Surface constraints versus Underlying Representation. Current trends in phonology: Models and methods, ed. by J. Durand and B. Laks, 97–122. CNRS, Paris, and University of Salford: University of Salford Publications.

Kenstowicz, Michael. 1997. Base Identity and Uniform Exponence: Alternatives to Cyclicity. Current Trends in Phonology: Models and Methods, ed. by Jacques Durand and Bernard Laks, 363–394. Salford: University of Salford.

References, cont.

Kenstowicz, Michael.; and Charles Kisseberth. 1977. Topics in Phonological Theory. New York: Academic Press.

Ko, Heejeong. 2006. Base-Output Correspondence in Korean Nominal Inflection. Journal of East Asian Linguistics 15.195–243.

McCarthy, John.; and Alan Prince. 1995. Faithfulness and Reduplicative Identity. UMOP 18, ed. by Jill N. Beckman, Laura Walsh Dickey, and Susanne Urbanczyk, 249–384. Amherst, MA: GLSA.